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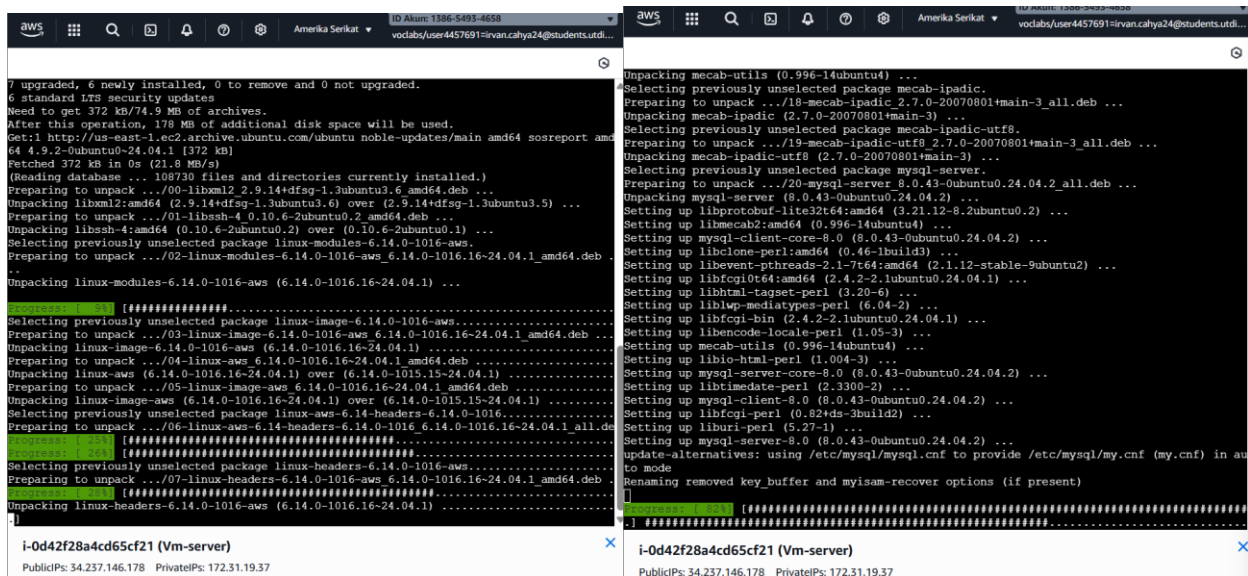
Rangkuman praktikum system terdesentralisasi dan terdistribusi

Implementasi Sistem Terdistribusi Sederhana dengan MySQL Antar VM di AWS EC2

Aturan masuk 4 Hapus

ID aturan grup keamanan sgr-0949a890e6e10b7a3	Jenis Info MYSQL/Aurora ▼	Protokol Info TCP
Rentang port Info 3306	Jenis sumber Info Kustom ▼	Sumber Info Q 0.0.0.0/0 ✕
Deskripsi - opsional Info 		

Pembahasan : Membuat vm server dan vm client , aturan masuk yang diedit adalah vm server saja dengan jenis MYSQL / Aurora dengan sumber 0.0.0.0/0 agar bisa di hubungi ip mana saja



```
ubuntu@ip-172-31-19-37:~$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-
-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
ubuntu@ip-172-31-19-37:~$
```

i-0d42f28a4cd65cf21 (Vm-server)

PublicIPs: 34.237.146.178 PrivateIPs: 172.31.19.37

Pembahasan : sudo apt update && sudo apt upgrade -y agar bisa install mysql di vm server berikut ini adalh perintahnya **sudo apt install mysql-server -y -> sudo systemctl enable mysql -> sudo systemctl start mysql** , selssai

```
bind-address          = 0.0.0.0
mysqlx-bind-address   = 127.0.0.1
#
# * Fine Tuning
#
key_buffer_size       = 16M
# max_allowed_packet  = 64M

Q Where Was      M-W Next
F Forward
```

i-0d42f28a4cd65cf21 (Vm-server)

PublicIPs: 34.237.146.178 PrivateIPs: 172.31.19.37

```
ubuntu@ip-172-31-19-37:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

i-0d42f28a4cd65cf21 (Vm-server)

PublicIPs: 34.237.146.178 PrivateIPs: 172.31.19.37

```
welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE USER 'akbar'@'%' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> GRANT ALL PRIVILEGES ON *.* TO 'akbar'@'%' WITH GRANT OPTION;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)

mysql> █
```

i-0d42f28a4cd65cf21 (Vm-server)

PublicIPs: 34.237.146.178 PrivateIPs: 172.31.19.37

Pembahasan : Konfigurasi MySQL agar Bisa Diakses dari VM client masuk ke ; sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf ; cari baris bind-address = 127.0.0.1, diganti ip nya menjadi bind-address = 0.0.0.0 agar bisa diakses semua ; untuk simpan ctrl + o ; enter; ctrl x ; untuk keluar lalu restart sudo systemctl restart mysql. Selanjutnya membuat user untuk akses jarak jauh dengan cara -> sudo mysql -> CREATE USER 'akbar'@'%' IDENTIFIED BY 'password123'; GRANT ALL PRIVILEGES ON *.* TO 'akbar'@'%' WITH GRANT OPTION; FLUSH PRIVILEGES; -> jika muncul terakhir mysql> berhasil

```
ubuntu@ip-172-31-24-75:~$ sudo apt install mysql-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  mysql-client-8.0 mysql-client-core-8.0 mysql-common
The following NEW packages will be installed:
  mysql-client mysql-client-8.0 mysql-client-core-8.0 mysql-common
0 upgraded, 4 newly installed, 0 to remove and 6 not upgraded.
Need to get 2779 kB of archives.
After this operation, 61.8 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-
core-8.0 amd64 8.0.43-0ubuntu0.24.04.2 [2740 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+
1.1.0build1 [6746 B]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-
8.0 amd64 8.0.43-0ubuntu0.24.04.2 [22.4 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client
all 8.0.43-0ubuntu0.24.04.2 [9404 B]
Fetched 2779 kB in 0s (51.0 MB/s)
Selecting previously unselected package mysql-client-core-8.0.
(Reading database ... 103371 files and directories currently installed.)
Preparing to unpack .../mysql-client-core-8.0_8.0.43-0ubuntu0.24.04.2_amd64.deb ...
Unpacking mysql-client-core-8.0 (8.0.43-0ubuntu0.24.04.2) ...
Selecting previously unselected package mysql-common.
Preparing to unpack .../mysql-common_5.8+1.1.0build1_all.deb ...
Unpacking mysql-common (5.8+1.1.0build1) ...
Selecting previously unselected package mysql-client-8.0.
Preparing to unpack .../mysql-client-8.0_8.0.43-0ubuntu0.24.04.2_amd64.deb ...
Unpacking mysql-client-8.0 (8.0.43-0ubuntu0.24.04.2) ...
Selecting previously unselected package mysql-client.
Preparing to unpack .../mysql-client_8.0.43-0ubuntu0.24.04.2_all.deb ...
Unpacking mysql-client (8.0.43-0ubuntu0.24.04.2) ...
```

i-000a8ff1c561f54d3 (Vm-client)

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75

```
ubuntu@ip-172-31-24-75:~$ mysql -h 172.31.19.37 -u akbar -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
[1]+  Stopped                  mysql -h 172.31.19.37 -u akbar -p
ubuntu@ip-172-31-24-75:~$ mysql -h 172.31.19.37 -u akbar -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> 
```

i-000a8ff1c561f54d3 (Vm-client)

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75



```
aws | ID Akun: 1386-5495-4658 | Amerika Serikat | voclabs/user4457691=irvan.cahya24@students.utdi...

owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE mahasiswa;
Query OK, 1 row affected (0.03 sec)

mysql> USE mahasiswa;
Database changed
mysql> CREATE TABLE mahasiswa (
  ->     nim VARCHAR(20) PRIMARY KEY,
  ->     nama VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO mahasiswa (nim, nama) VALUES
  ->     ('235410023', 'Agung'),
  ->     ('235410026', 'Budi'),
  ->     ('235410022', 'Sari');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM mahasiswa;
+-----+-----+
| nim      | nama  |
+-----+-----+
| 235410022 | Sari  |
| 235410023 | Agung |
| 235410026 | Budi  |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

i-000a8ff1c561f54d3 (Vm-client) ✕
PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75

Pembahasan : menguji koneksi dari vm client lakukan **sudo apt update -> sudo apt install mysql-client -y** agar terhubung dengan server kita catat ip server yang dibuat 172.31.19.37 lalu jalankan perintah -> **mysql -h 172.31.19.37 -u akbar -p** -> jika sudah muncul mysql> berhasil. Selanjutnya melakukan query antar vm perintah:

CREATE DATABASE mahasiswa;

USE mahasiswa;

CREATE TABLE mahasiswa (

 nim VARCHAR(20) PRIMARY KEY,

 nama VARCHAR(100)

);

```
INSERT INTO mahasiswa (nim, nama) VALUES
```

```
('235410023', 'Agung'),
```

```
('235410026', 'Budi'),
```

```
('235410022', 'Sari');
```

```
SELECT * FROM mahasiswa;
```

selesai

Kesimpulan : Jadi pada praktikum ini kita membuat query sql di client tapi datanya tersimpan di server karena mysql -h 172.31.30.198 -u akbar -pyang menghubungkan kedua vm